



Ranga Reddy

Senior Data Engineer | Open Source Contributor (Apache Hudi, XTable) | Ex-Spark & Iceberg Backline Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with **14+ years** of experience building petabyte-scale Big Data platforms and lakehouse architectures. Currently contributing as an **Open Source Engineer** at OneHouse, with upstream commits to **Apache Hudi** and **Apache XTable**—designing core components for indexing, compaction, clustering, and multi-writer concurrency.

Previously served as Cloudera's **Spark & Iceberg Backline Engineer**, serving as the final escalation point for the platform's most critical P0/P1 customer issues across Fortune 500 deployments. Deeply specialized in **Spark performance tuning**, **Iceberg table optimization**, distributed systems root-cause analysis, and production troubleshooting across Spark, Trino, Presto, and Flink ecosystems.

Core Strengths:

- Open Source Development & Community Leadership (Apache Hudi, XTable)
- **Spark & Iceberg Backline Engineering** – Production Escalation & Root-Cause Analysis
- Spark SQL & DataFrame Optimization · Catalyst Query Tuning
- Lakehouse Architecture (Hudi, Iceberg, Hive Metastore)
- Real-time & Batch Data Pipeline Engineering
- Customer-facing Solution Engineering & Technical Content Creation

EDUCATION

- **Master of Computer Applications (MCA)** – Sri Krishnadevaraya University, Anantapur | 2011 | 80%
- **Bachelor of Science (B.Sc)** – Sri Venkateswara University, Tirupati | 2008 | 81%

CORE SKILLS

Domain	Technologies
Lakehouse / Table Formats	Apache Hudi (Contributor), Apache Iceberg, Hive Metastore
Distributed Compute	Apache Spark (SQL, Streaming, DataFrames), Apache Storm,

	Apache Hadoop
Streaming & Messaging	Apache Kafka, Hudi Streamer (DeltaStreamer)
Query Engines	Trino, PrestoDB, Athena, Spark SQL, Hive
Cloud & Storage	AWS (S3, Athena, Glue, EMR), GCP, MinIO
Languages	Java, Scala, Python
Frameworks	Spring, Hibernate
Databases	Oracle, MySQL, HBase
Distributions	Cloudera CDP/CDH, Hortonworks HDP, Onehouse
Tooling	Maven, Gradle, Jenkins, Git, IntelliJ

PROFESSIONAL EXPERIENCE

Open Source Engineer | OneHouse, Bangalore

Sep 2024 – Present

Contributing upstream to Apache Hudi and Apache XTable, designing and optimizing core components for large-scale data lakehouse systems while improving performance, scalability, and reliability across distributed environments.

Key Contributions:

- **Upstream Development:** Authored and reviewed pull requests, JIRAs, and RFCs across Hudi internals—indexing strategies, compaction, clustering, metadata table optimization, and multi-writer concurrency models.
- **Customer Escalation Engineering:** Lead escalation engineer for community + enterprise customer issues filed via Slack, GitHub, and Hudi mailing list; consistently resolve complex issues spanning Spark, Trino, Presto, and Flink integrations.
- **Benchmarking Framework:** Designed automated performance and correctness validation suites comparing Hudi behavior across Spark, Presto, and Trino—used internally to gate releases.
- **Customer Cost-Saving Tooling:** Built end-to-end tooling that parses table metadata, query logs, and statistics to generate actionable metrics; **enabled customers to save up to 40% on production costs.**
- **Technical Content:** Publish deep-dive blogs and tutorials on Hudi internals (indexing strategies, compaction tuning, metadata table optimization) for the OSS community.

Tech Stack: Java, Scala, Python, Spark, Trino, Presto, AWS(EMR, S3, Athena, Glue), GCP(Dataproc, GCS, Bigquery)

Staff Software Engineer (Spark & Iceberg Backline) | Cloudera, Bangalore

Mar 2020 – Aug 2024 · 4.5 years

Served as the final escalation point for Cloudera's global support organization—resolved the most complex Spark and Iceberg production outages that frontline support and field engineers could not.

Key Contributions:

- **Spark Production Escalations:** Led root-cause analysis on hundreds of P0/P1 Spark issues across Cloudera CDP/CDH deployments—diagnosing shuffle failures, OOM patterns, dynamic allocation regressions, Catalyst optimizer edge cases, and Hive-on-Spark / Spark-on-YARN integration breaks.
- **Iceberg Production Support:** Diagnosed and resolved critical production outages related to **Apache Iceberg** metadata corruption, snapshot isolation conflicts, time-travel query regressions, and partition/schema evolution compatibility across Spark 3.x and Hive Metastore. Routinely debugged Iceberg's integration with Trino and Presto for enterprise customers.
- **Internal Tooling:** Built a unified log-parsing tool that ingests Spark event logs, application logs, **and Iceberg metadata files** to surface stage-level bottlenecks and table-level inefficiencies; adopted by the global support team, drastically reducing average issue resolution time.
- **Cross-Platform Debugging:** Resolved complex integration issues between Spark, Iceberg, and cloud storage (AWS S3, GCS), including consistency model challenges and file listing performance bottlenecks.
- **Knowledge Sharing & Training:** Authored and published Spark & Iceberg knowledge articles on the Cloudera Community. Designed and ran "**Spark Hogwarts**" —a hands-on internal training program upskilling the Frontline support team on Spark internals and modern lakehouse formats (Iceberg/Hudi), reducing escalation rates significantly.

Tech Stack: Java, Scala, Python, Spark(all 2.x and 3.x), Kafka, Iceberg, Hive, Cloudera distributions (CDP,CDH,HDP) & its tools

Key Achievement: Enhanced Cloudera support documentation and tooling ecosystem, improving customer experience and troubleshooting speed.

Big Data Engineer | Dell EMC, Bangalore

Nov 2015 – Mar 2020 · 4.5 years

Project: Support Assist Intelligence Engine (SAIE)

Delivered a solution to store data from Support Assist, enabling reporting, support, and information exchange based on customer environments—facilitating a proactive and predictive support environment.

Role and Contributions:

- **Real-Time Pipeline:** Designed Spark → Kafka → Storm → HBase ingest path handling unstructured and semi-structured telemetry at scale.
- **Data Processing:** Read data from Oracle, processed it through Spark, and stored it in Kafka; read data from Kafka queues using Storm for real-time analysis.

- **Cluster Upgrade:** Co-led HDP 2.3 → 2.5 production upgrade across the 40-node cluster with zero unplanned downtime.
- **Storage Optimization:** Stored large volumes of processed data into HBase, ensuring efficient storage and retrieval for further analysis.
- **Deployment Support:** Supported Production Engineering team during deployment activities and interacted with Jenkins and Build Management systems.

Tech Stack: Hadoop, Hortonworks (HDP 2.6.5), Spark, Storm, Kafka, Oracle, HBase, JSON

Project: SAIE Analytics

Extracted server data and published to OMNEO UI for analysis, providing actionable insights to the Analytics team.

Role and Contributions:

- **End-to-End Design:** Led the design of the SAE (Support Assist Enterprise) module, contributing to overall architecture and functionality.
- **Data Integration:** Mapped attributes and objects to ensure seamless integration across different SAE releases, ensuring data consistency and compatibility.
- **Ingestion Pipeline:** Designed and implemented the ingestion process for configuration files used as reference mapping in the SAE module.
- **Data Transformation:** Used Spark to read JSON data from Oracle, process it, and generate CSV outputs for seamless data extraction.

Tech Stack: Java, Oracle DB, Hortonworks, Hadoop, Spark, JSON, XML

Key Achievement: Received **Dell Silver Award (Q4 FY18)** and **Cash Award** for exemplary performance on SAIE projects.

Senior Software Engineer | Mindtree Pvt. Ltd, Bangalore

Aug 2014 – Nov 2015 · 1.4 years

Project: Meeting Services Application (MSA)

Built a mobile-and-tablet platform enabling meeting planners to request services without leaving the room.

Role and Contributions:

- Implemented change requests provided by clients.
- Designed and developed Controller, Service, and DAO layer classes establishing application structure and logic.
- Developed Spring Service Interfaces and implementations for modular service operations.
- Designed JSP pages using jQuery and AngularJS for interactive user interfaces.
- Used Hibernate for streamlined database operations and Log4j for effective logging and monitoring.

Tech Stack: Java, Spring, Hibernate, AngularJS, JQuery, HTML, MySQL

Key Achievement: Awarded "**A-Team Performer**" for excellence in project delivery.

Junior Software Engineer | IApp Software Pvt. Ltd, Bangalore

Oct 2012 – Jul 2014 · 1.10 years

Project: AppCenter Cloud Management Platform

A unified cloud services delivery and business management platform enabling cloud services automation for provisioning, lifecycle, monitoring, and user management.

Role and Contributions:

- Used Spring framework for dependency injection and component integration.
- Created Portlets using Spring Portlet MVC for dynamic web components.
- Generated Controller Classes for request mapping and business logic.
- Designed JSP pages with visually appealing and user-friendly interfaces.
- Used Kendo UI for JavaScript code and client-side validation.
- Conducted application testing using JUnit.

Tech Stack: Java, Spring, RDB, Hibernate, MySQL, Kendo UI, JQuery, HTML, CSS

OPEN SOURCE CONTRIBUTIONS

- **Apache Hudi:** Active contributor with upstream commits across indexing, compaction, clustering, metadata table, and multi-writer concurrency features.
- **Apache XTable:** Active contributor working on cross-platform lakehouse interoperability.
- **Benchmarking Framework:** Designed automated performance and correctness validation suites for Hudi across Spark, Presto, and Trino.
- **Customer Tooling:** Built end-to-end cost-saving tool for Hudi customers (up to 40% cost reduction).

ACHIEVEMENTS

- 🏆 **Silver Award** – Dell EMC (Q4 FY18).
- 💡 **Cash Award** – For contributions to SAIE Analytics project (Dell EMC).
- 🚀 **A-Team Award** – Best performer, MSA project (Mindtree).
- 💻 Active contributor to **Cloudera Community Articles**.

-  Participant in **Cloudera Hackathon** and **Dell Ideafest**, showcasing innovative tools and ideas.

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge and belief.

Place: Bangalore, India

Date:

(Ranga Reddy)